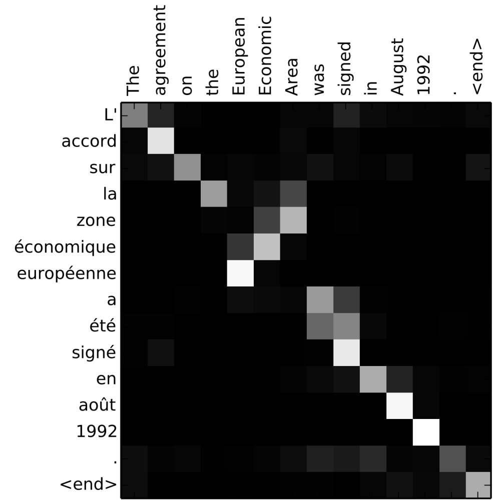
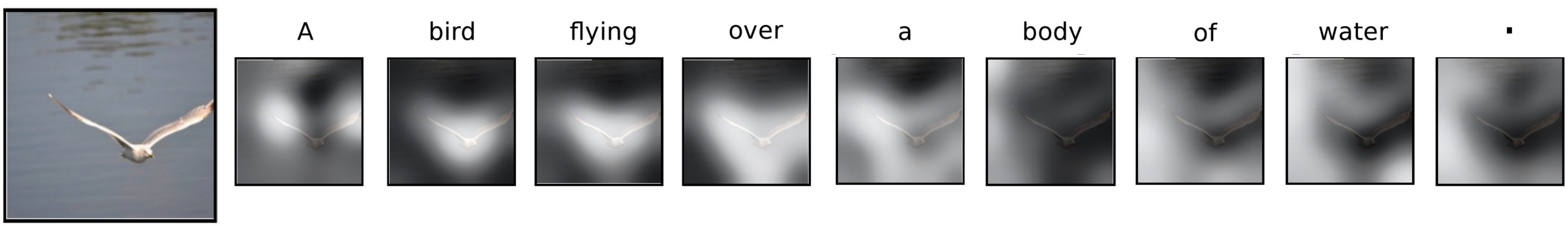
Sequence-to-sequence models with attention





Many slides adapted from [J. Johnson]

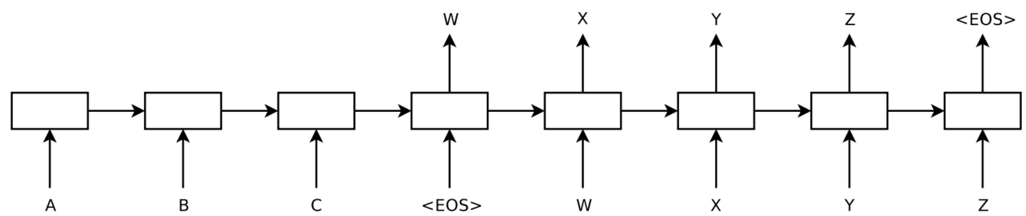
Outline

- Vanilla seq2seq with RNNs
- Seq2seq with RNNs and attention
- Image captioning with attention

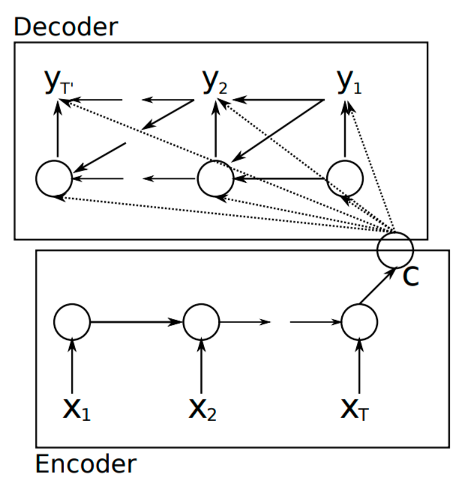
Sequence-to-sequence modeling: Machine translation

"We are eating bread" ➡ "Estamos comiendo pan"

Sequence-to-sequence modeling with RNNs

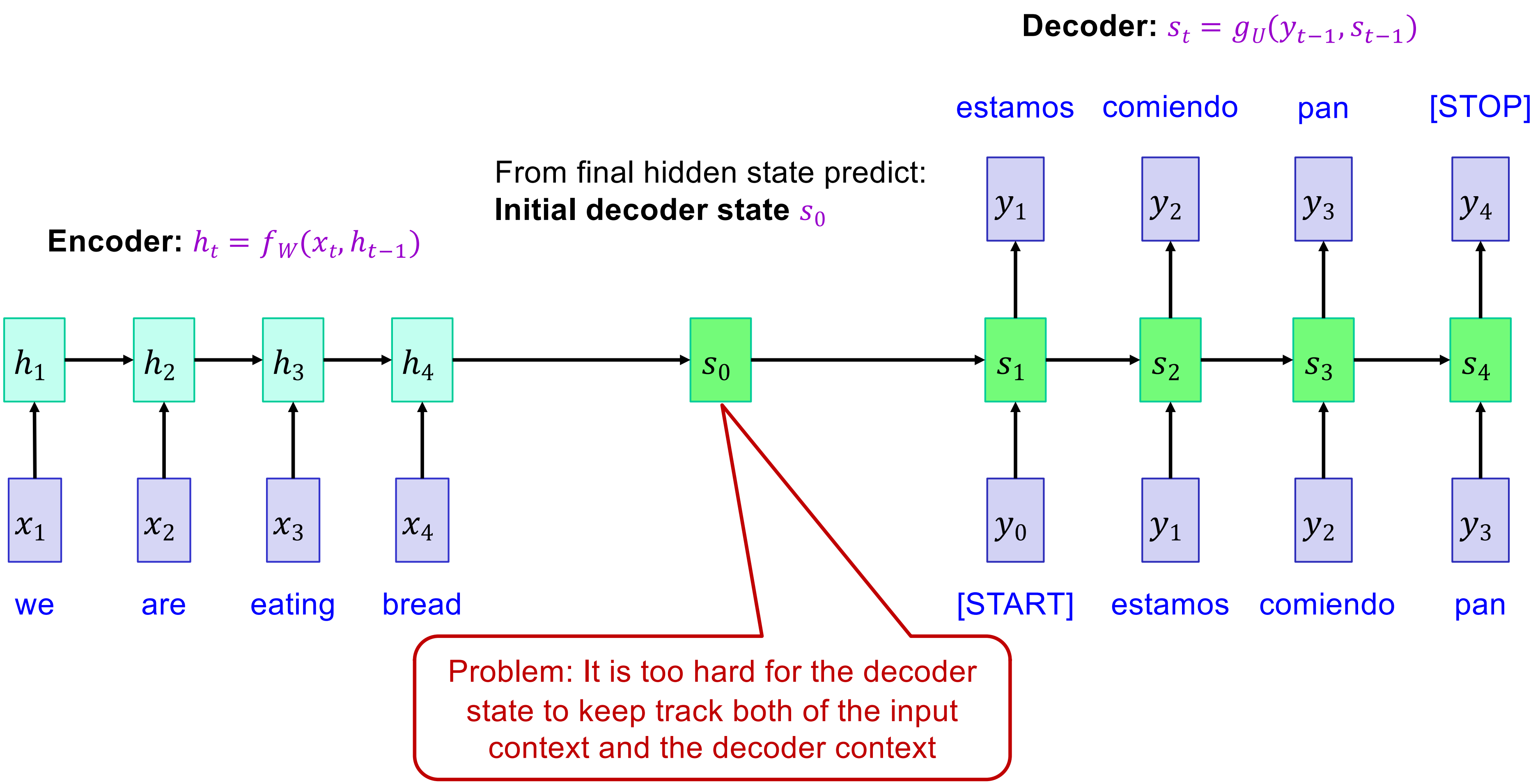


I. Sutskever, O. Vinyals, Q. Le, [Sequence to Sequence Learning with Neural Networks](), NeurIPS 2014



K. Cho, B. Merrienboer, C. Gulcehre, F. Bougares, H. Schwenk, and Y. Bengio, [Learning phrase representations using RNN encoder-decoder for statistical machine translation](), ACL 2014

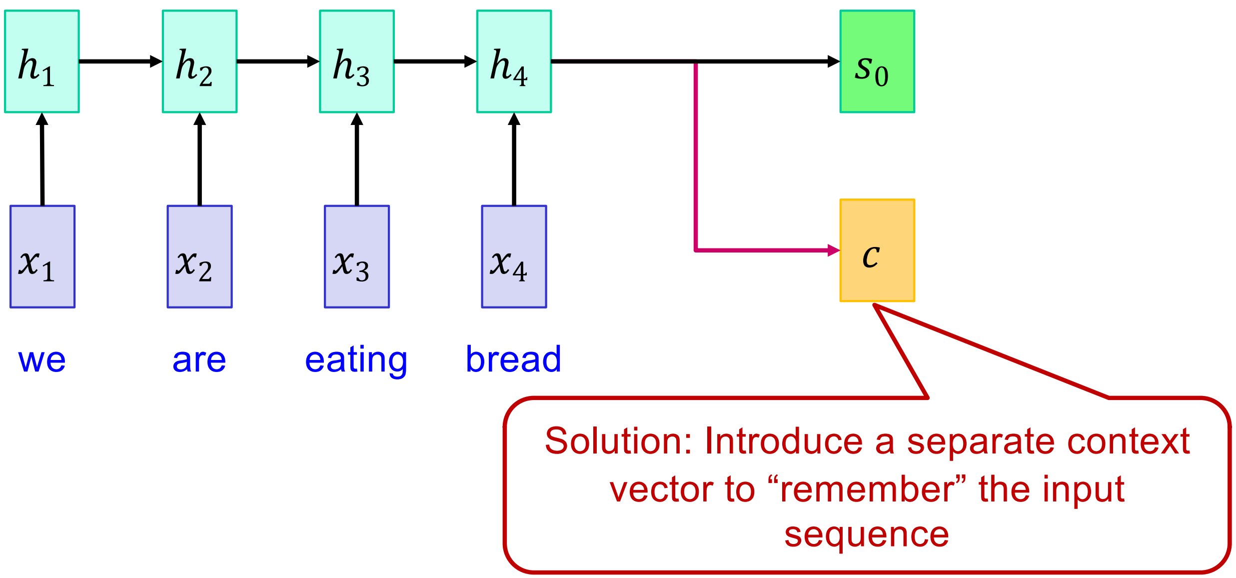
Sequence-to-sequence modeling with RNNs



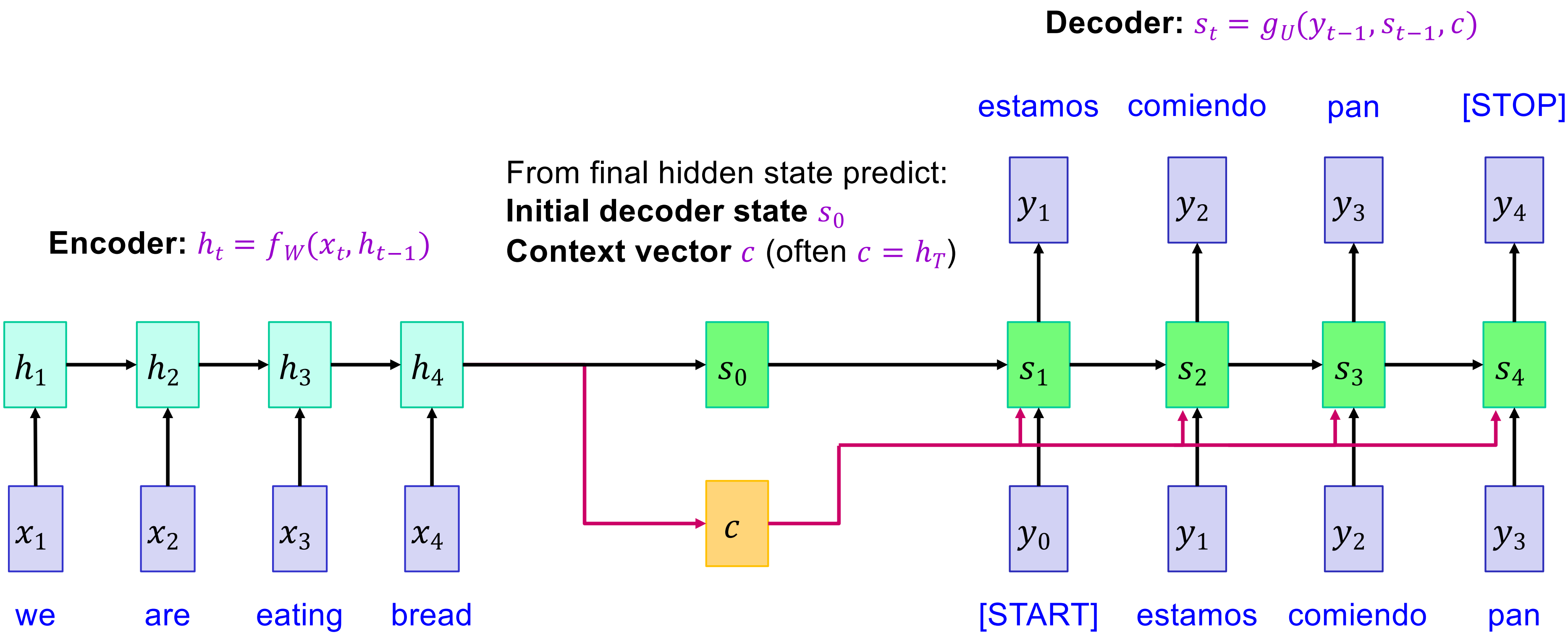
Sequence-to-sequence modeling with RNNs

From final hidden state predict:
Initial decoder state s_0
Context vector c (often $c = h_T$)

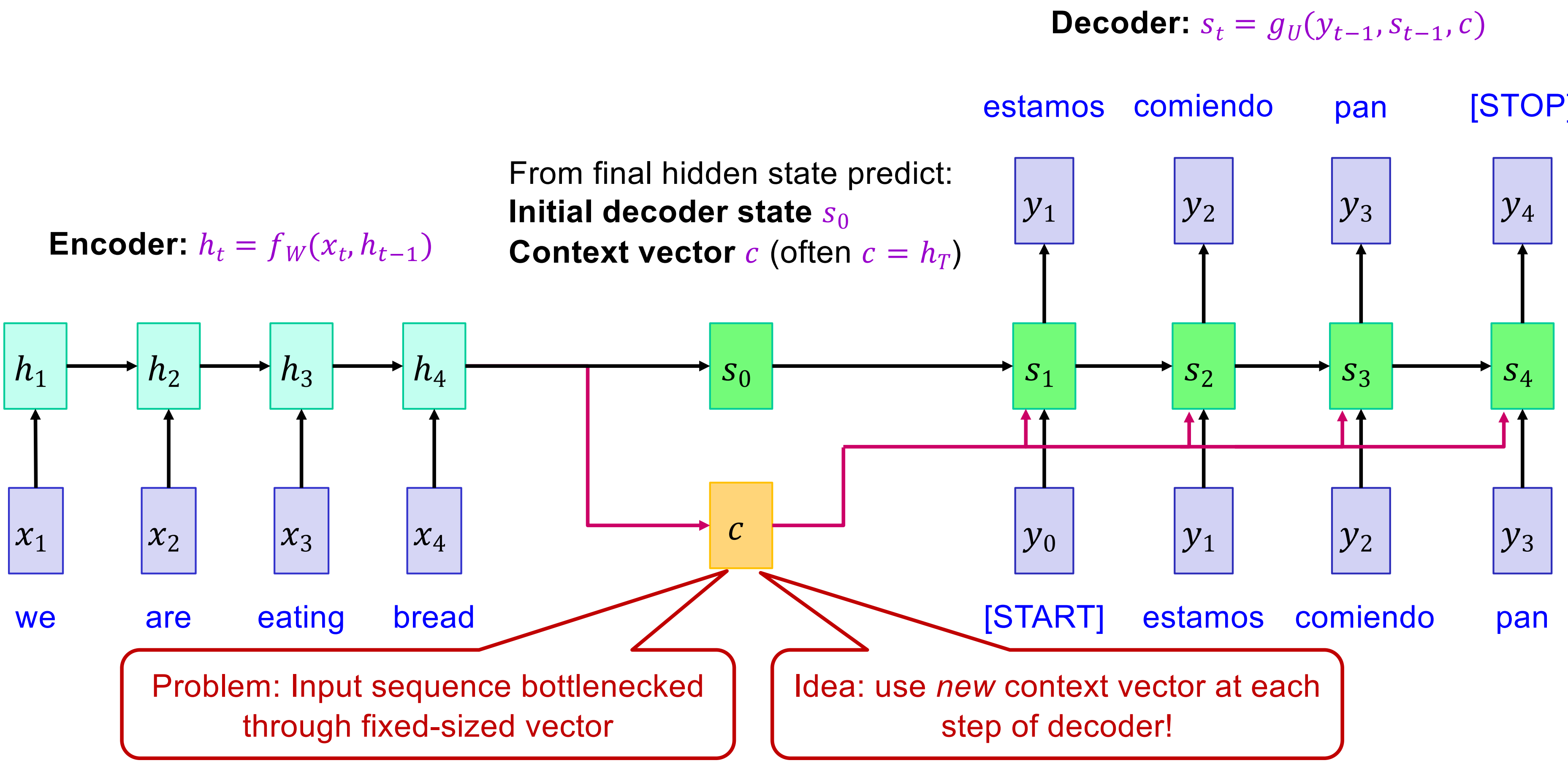
Encoder: $h_t = f_W(x_t, h_{t-1})$



Sequence-to-sequence modeling with RNNs

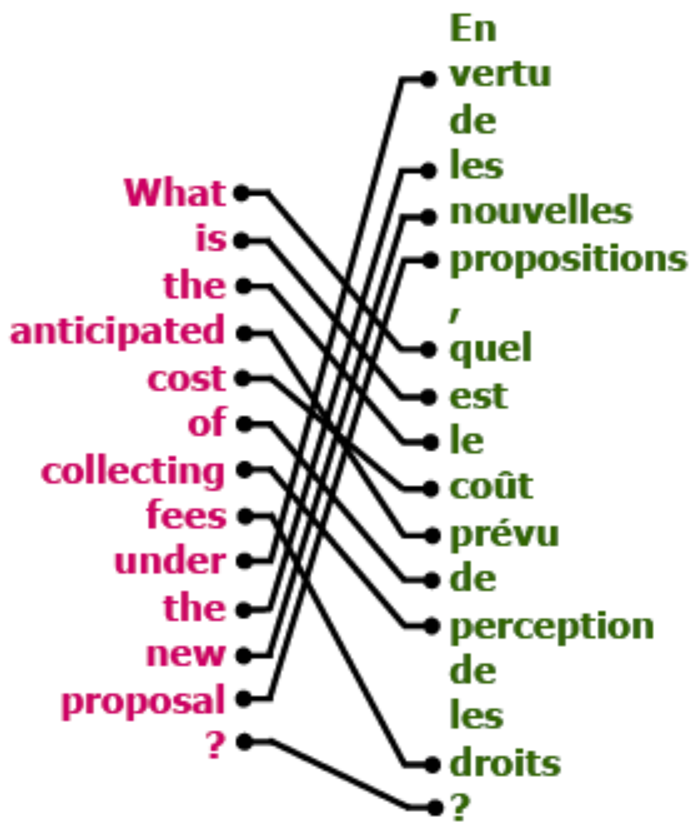


Sequence-to-sequence modeling with RNNs



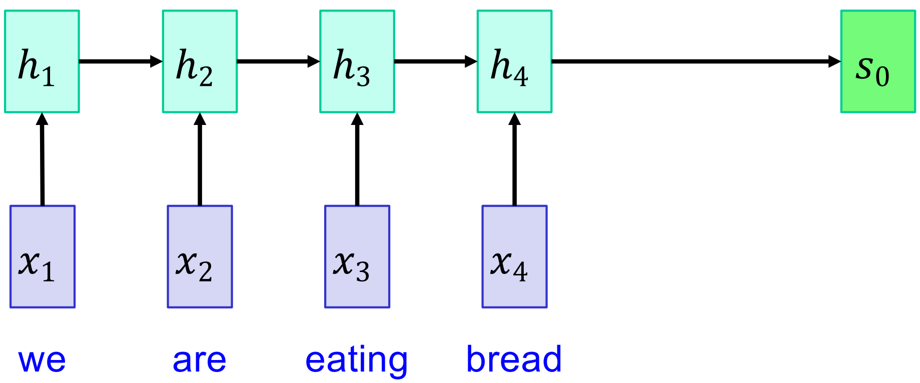
Sequence-to-sequence with RNNs *and attention*

- Intuition: translation requires *alignment*

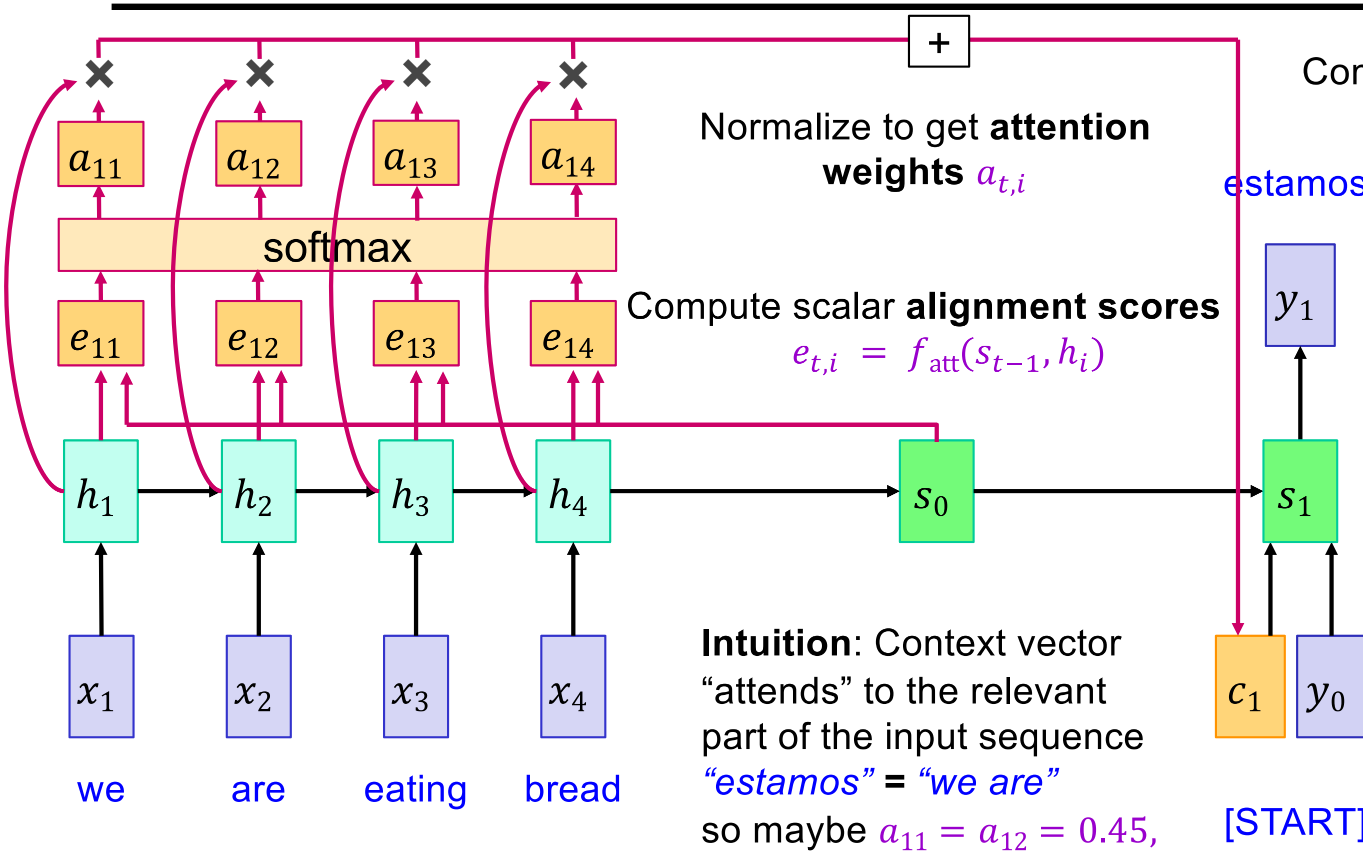


Sequence-to-sequence with RNNs and attention

- At each timestep of decoder, context vector “looks at” different parts of the input sequence



Sequence-to-sequence with RNNs and attention



Normalize to get **attention weights** $a_{t,i}$

Compute scalar **alignment scores**

$$e_{t,i} = f_{\text{att}}(s_{t-1}, h_i)$$

Compute **context vector** as

$$c_t = \sum_i a_{t,i} h_i$$

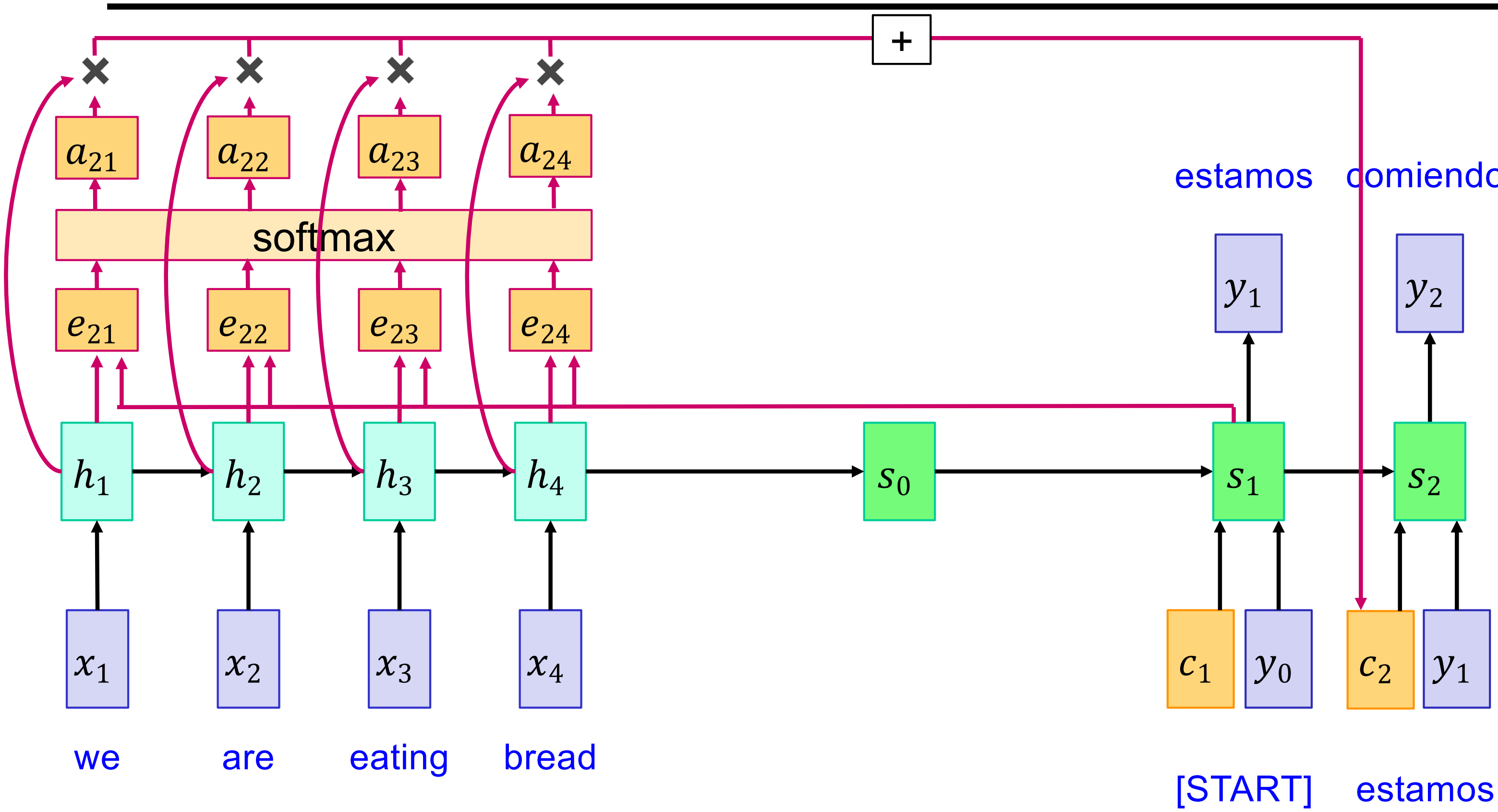
Use context vector in decoder:

$$s_t = g_U(y_{t-1}, s_{t-1}, c_t)$$

Intuition: Context vector "attends" to the relevant part of the input sequence
"estamos" **=** *"we are"*
so maybe $a_{11} = a_{12} = 0.45$,
$a_{13} = a_{14} = 0.05$

D. Bahdanau, K. Cho, Y. Bengio, Neural Machine Translation by Jointly Learning to Align and Translate, ICLR 2015

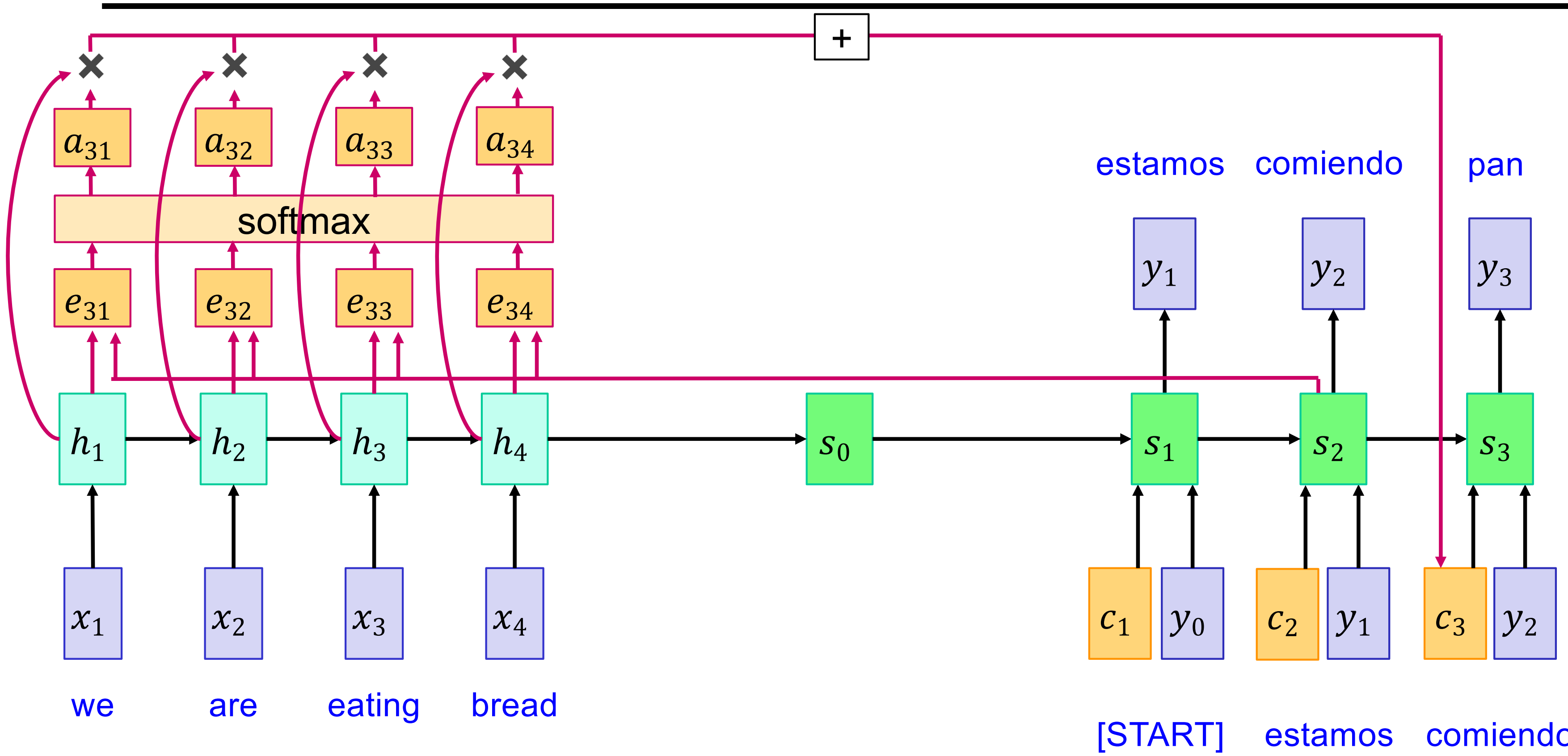
Sequence-to-sequence with RNNs and attention



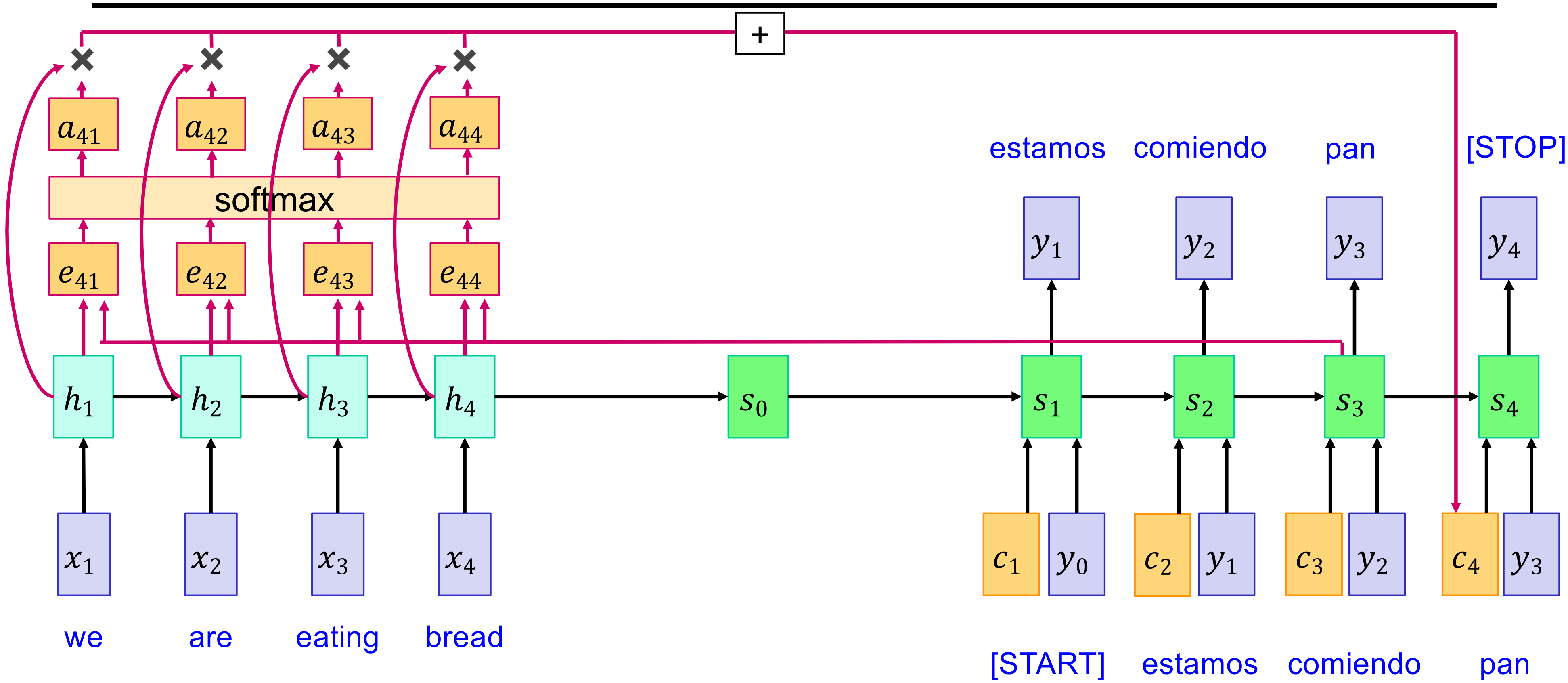
Use c_2 to compute s_2, y_2

Repeat: Use s_1 to compute new context vector c_2

Sequence-to-sequence with RNNs and attention



Sequence-to-sequence with RNNs and attention



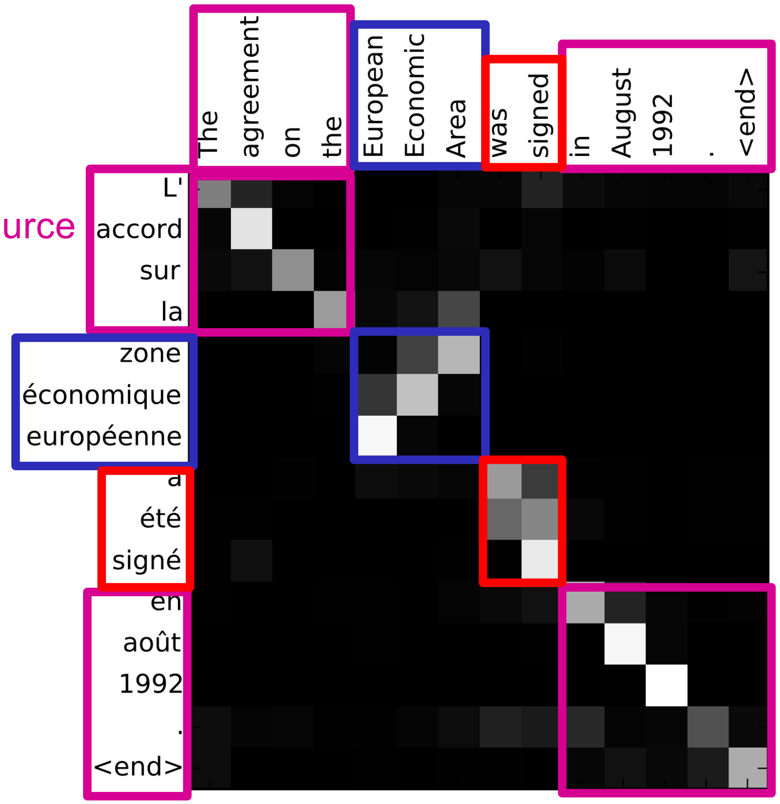
Sequence-to-sequence with RNNs and attention

- Visualizing attention weights (English source, French target):

Same word order in source and target languages

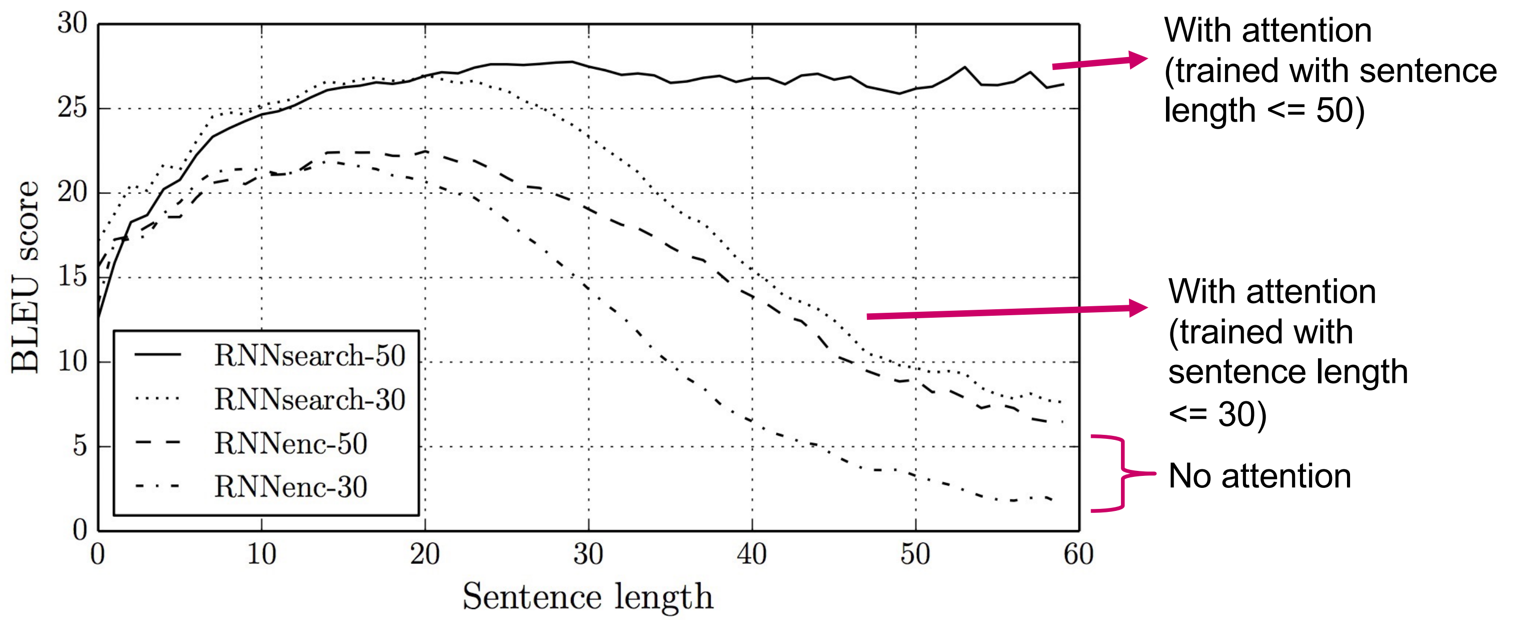
Flipped word order

Verb conjugation is different



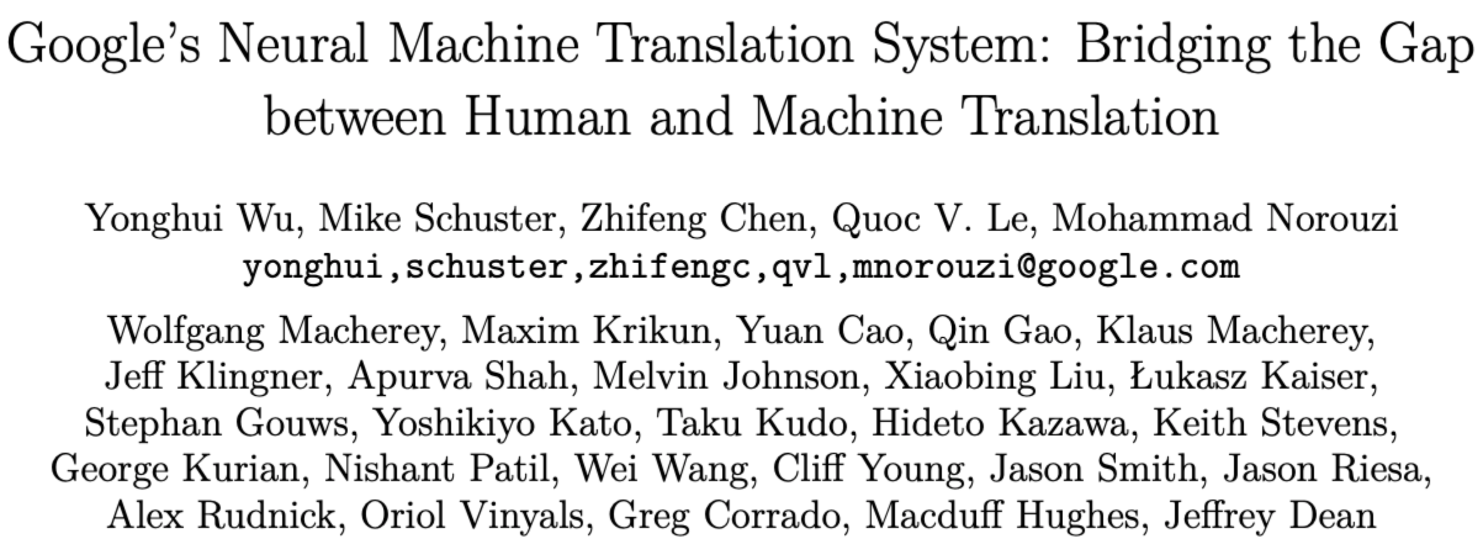
D. Bahdanau, K. Cho, Y. Bengio, [Neural Machine Translation by Jointly Learning to Align and Translate](), ICLR 2015

Quantitative evaluation



D. Bahdanau, K. Cho, Y. Bengio, [Neural Machine Translation by Jointly Learning to Align and Translate](), ICLR 2015

Google Neural Machine Translation (GNMT)


Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

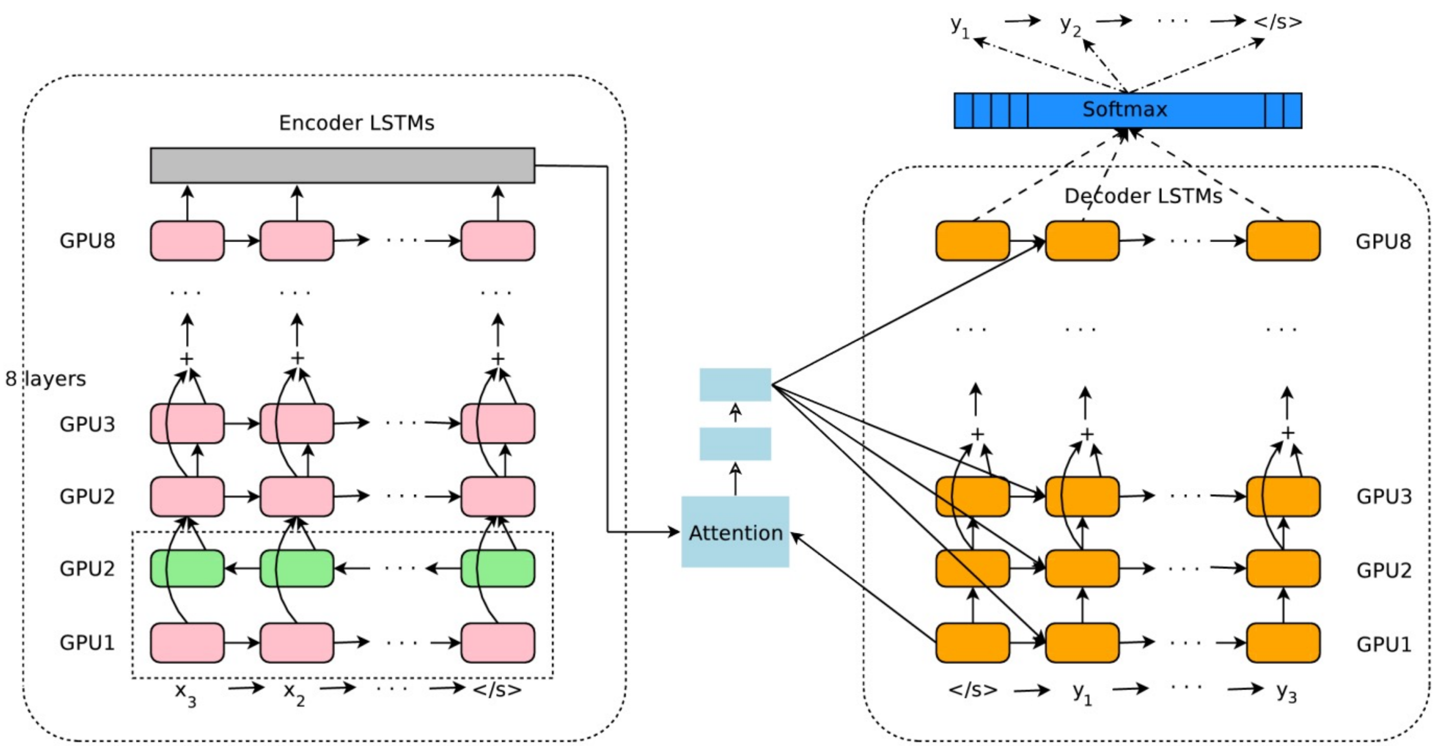
Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi
yonghui,schuster,zhifengc,qvl,mnorouzi@google.com

Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey,
Jeff Klingner, Apurva Shah, Melvin Johnson, Xiaobing Liu, Łukasz Kaiser,
Stephan Gouws, Yoshikiyo Kato, Taku Kudo, Hideto Kazawa, Keith Stevens,
George Kurian, Nishant Patil, Wei Wang, Cliff Young, Jason Smith, Jason Riesa,
Alex Rudnick, Oriol Vinyals, Greg Corrado, Macduff Hughes, Jeffrey Dean

Y. Wu et al., Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation, arXiv 2016

https://www.nytimes.com/2016/12/14/magazine/the-great-ai-awakening.html

Google Neural Machine Translation (GNMT)



Y. Wu et al., [Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation], arXiv 2016

Google Neural Machine Translation (GNMT)

- **Standard training objective:** maximize log-likelihood of ground truth output given input:

$$\sum_i \log P_W(Y_i^* | X_i)$$

  - Only encourages the system to reproduce the reference sentences, does not induce a very good ranking on outputs that don't match reference sentences
  - Not related to task-specific evaluation metric (e.g., BLEU score)
- **Refinement objective:** expectation of rewards over possible predicted sentences Y:

$$\sum_i \sum_Y P_W(Y | X_i)\, R(Y, Y_i^*)$$

  - Use variant of BLEU score to compute reward
  - Reward is not differentiable -- need RL to train (initialize with ML-trained model)

Google Neural Machine Translation (GNMT)

- Human evaluation results on production data (500 randomly sampled sentences from Wikipedia and news websites)

Table 10: Mean of side-by-side scores on production data

	PBMT	GNMT	Human	Relative Improvement
English $\rightarrow$ Spanish	4.885	5.428	5.550	87%
English $\rightarrow$ French	4.932	5.295	5.496	64%
English $\rightarrow$ Chinese	4.035	4.594	4.987	58%
Spanish $\rightarrow$ English	4.872	5.187	5.372	63%
French $\rightarrow$ English	5.046	5.343	5.404	83%
Chinese $\rightarrow$ English	3.694	4.263	4.636	60%

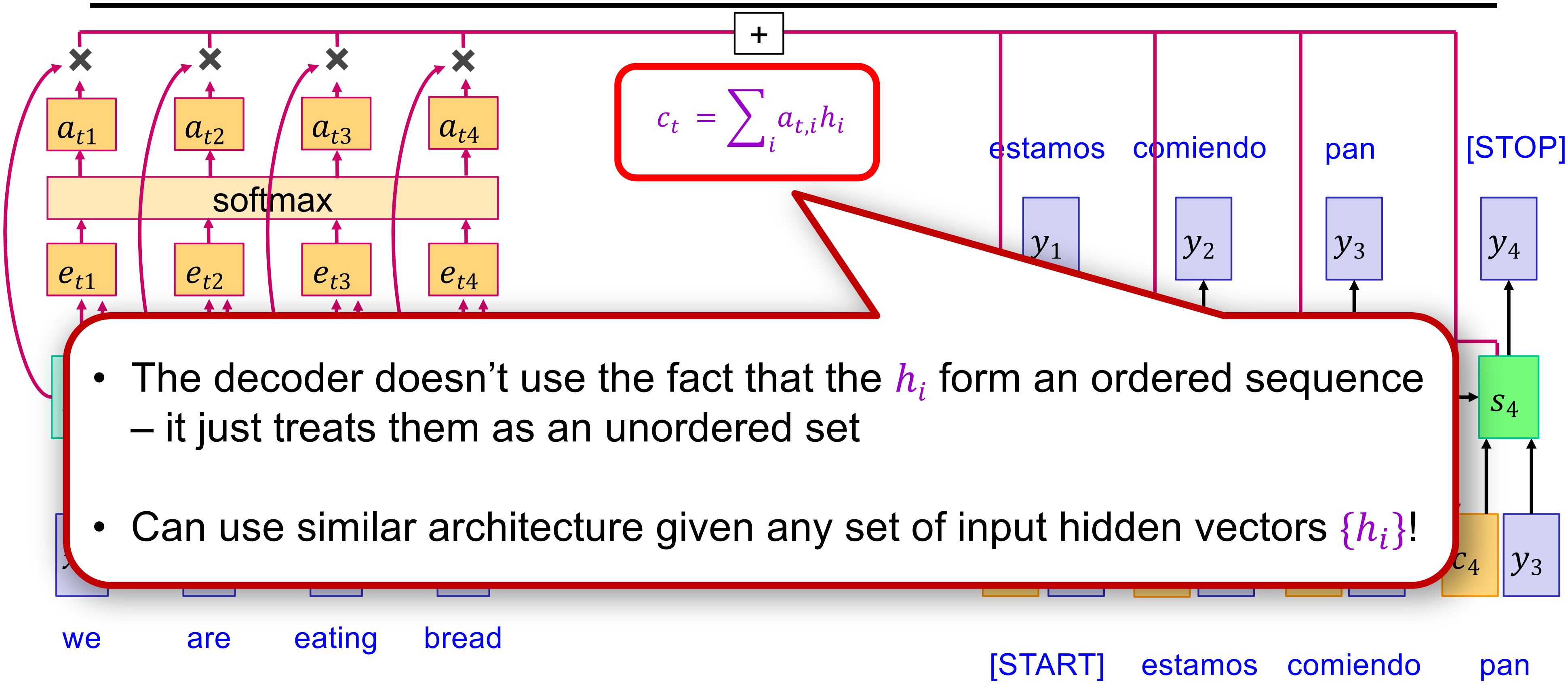
Side-by-side scores: range from 0 (“completely nonsense translation”) to 6 (“perfect translation”), produced by human raters fluent in both languages

PBMT: Translation by phrase-based statistical translation system used by Google
GNMT: Translation by GNMT system
Human: Translation by humans fluent in both languages

Outline

- Vanilla seq2seq with RNNs
- Seq2seq with RNNs and attention
- Image captioning with attention

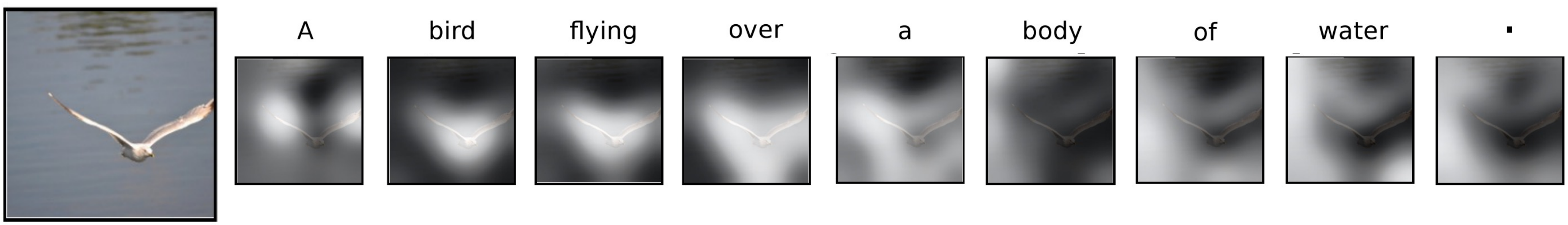
Generalizing attention



- The decoder doesn't use the fact that the h_i form an ordered sequence – it just treats them as an unordered set
- Can use similar architecture given any set of input hidden vectors $\{h_i\}$!

Image captioning with RNNs and attention

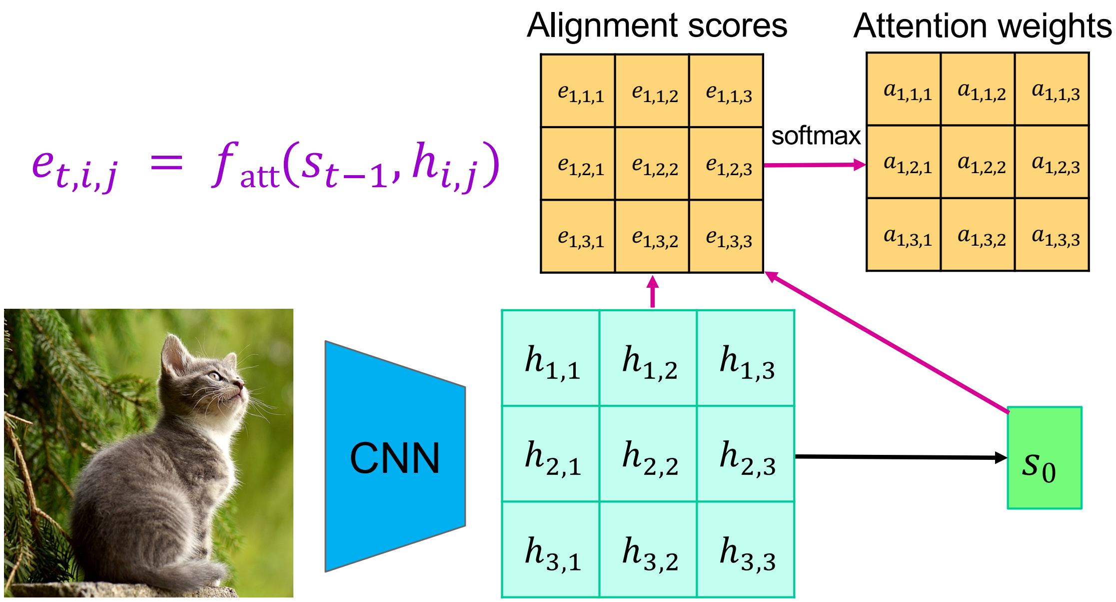
- Idea: pay attention to different parts of the image when generating different words
- Automatically learn this *grounding* of words to image regions without direct supervision



K. Xu et al., [Show, Attend and Tell: Neural Image Caption Generation with Visual Attention](), ICML 2015

Image captioning with RNNs and attention

$$e_{t,i,j} = f_{\text{att}}(s_{t-1}, h_{i,j})$$



Use CNN to extract a grid of features

K. Xu et al., [Show, Attend and Tell: Neural Image Caption Generation with Visual Attention](), ICML 2015

Image captioning with RNNs and attention

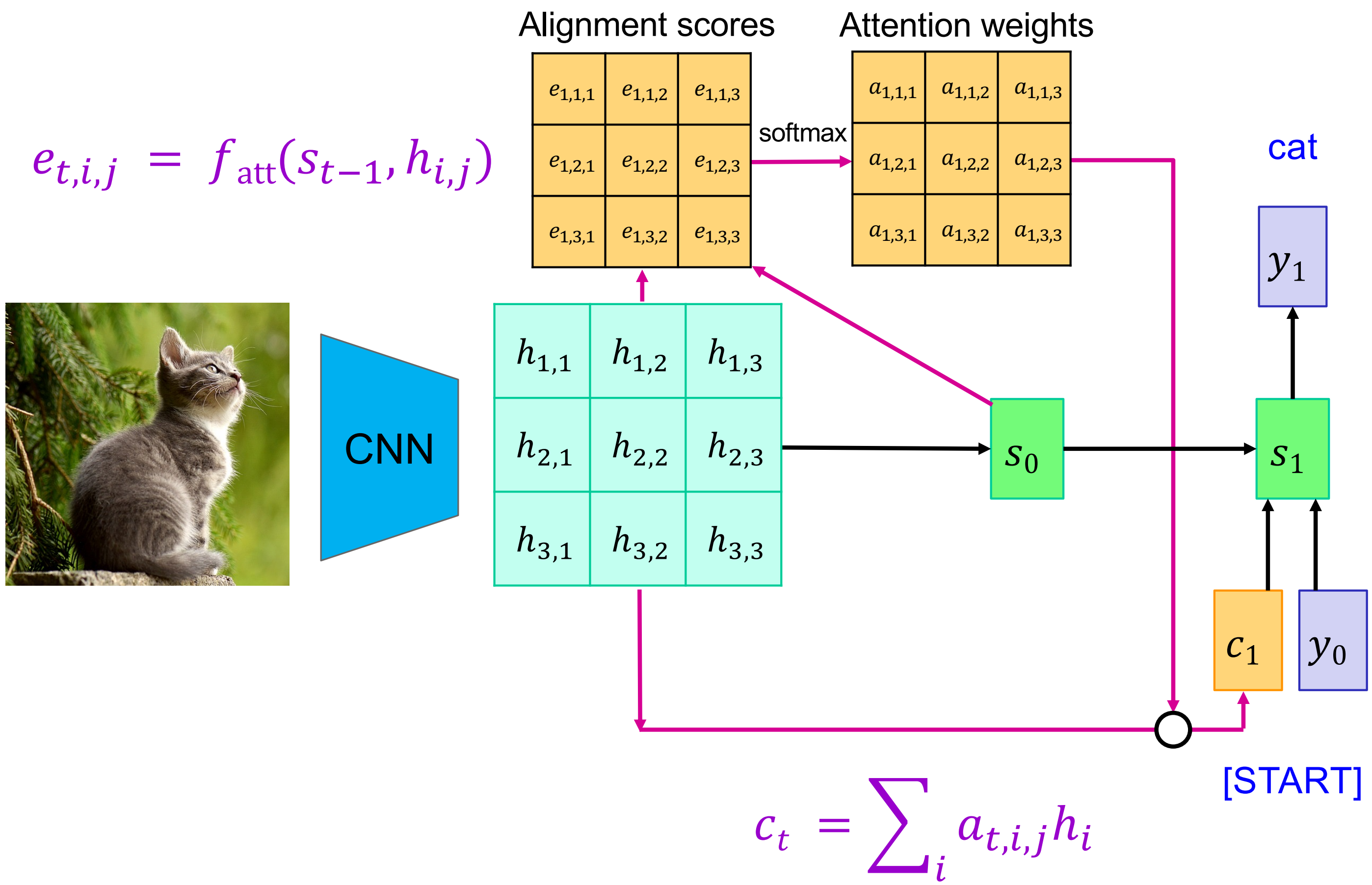
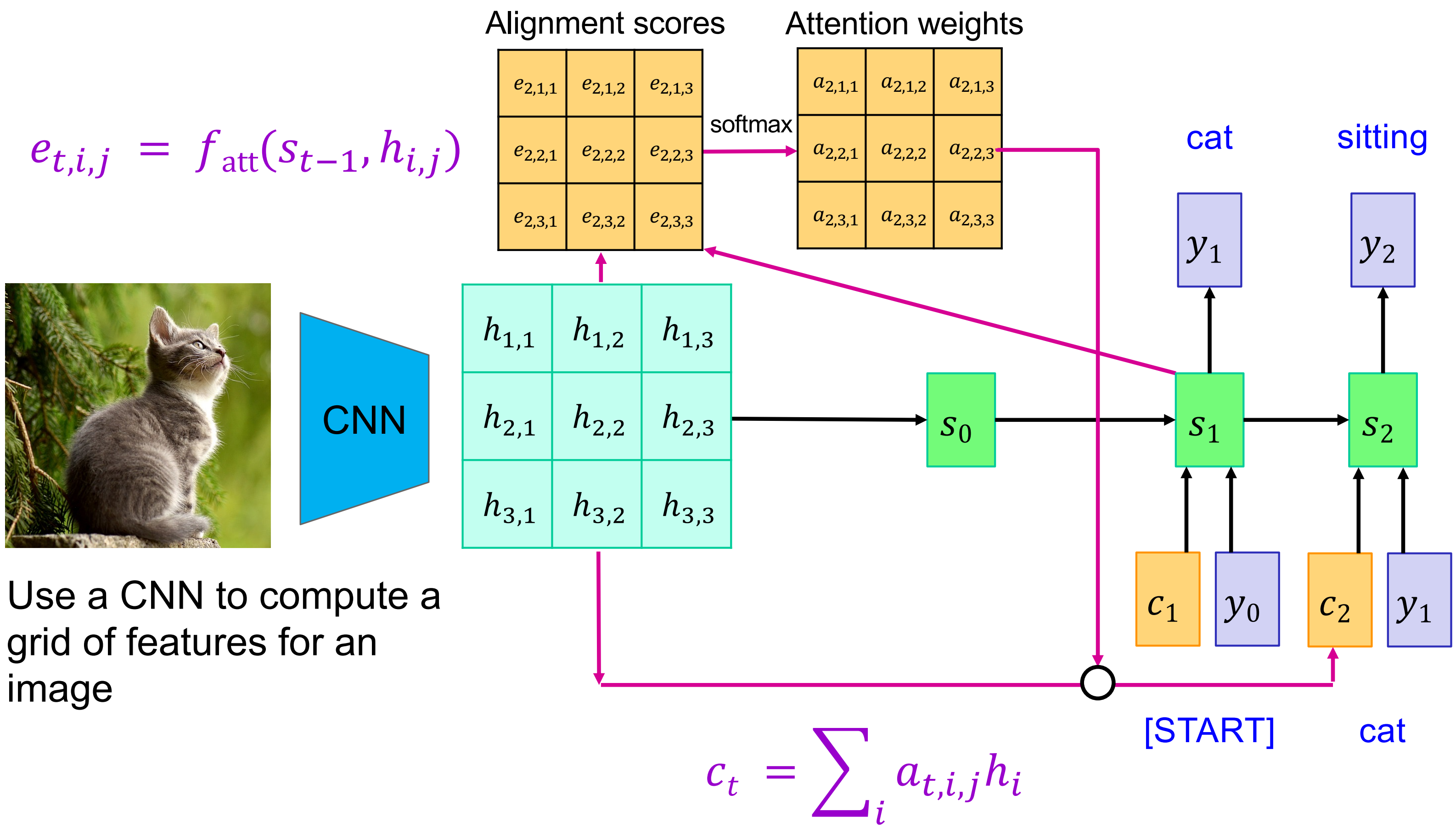
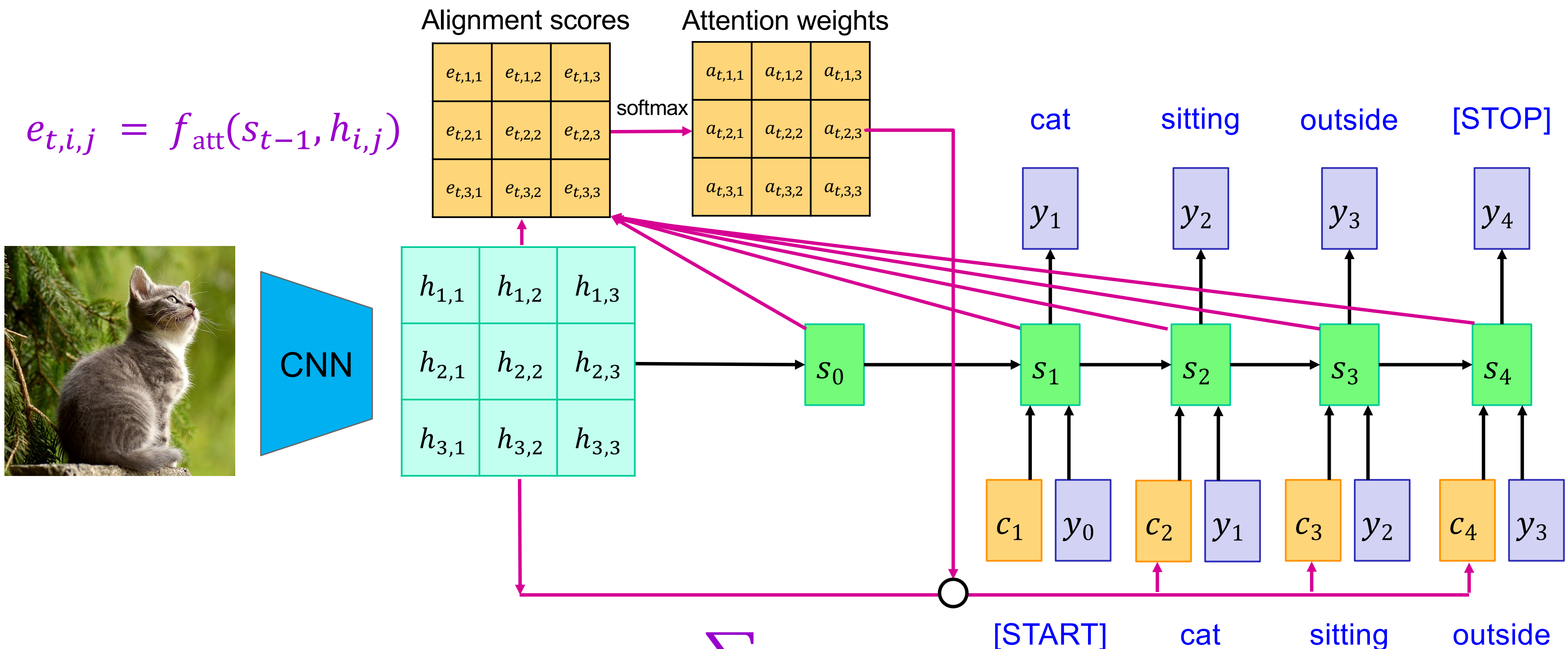


Image captioning with RNNs and attention



Use a CNN to compute a grid of features for an image

Image captioning with RNNs and attention



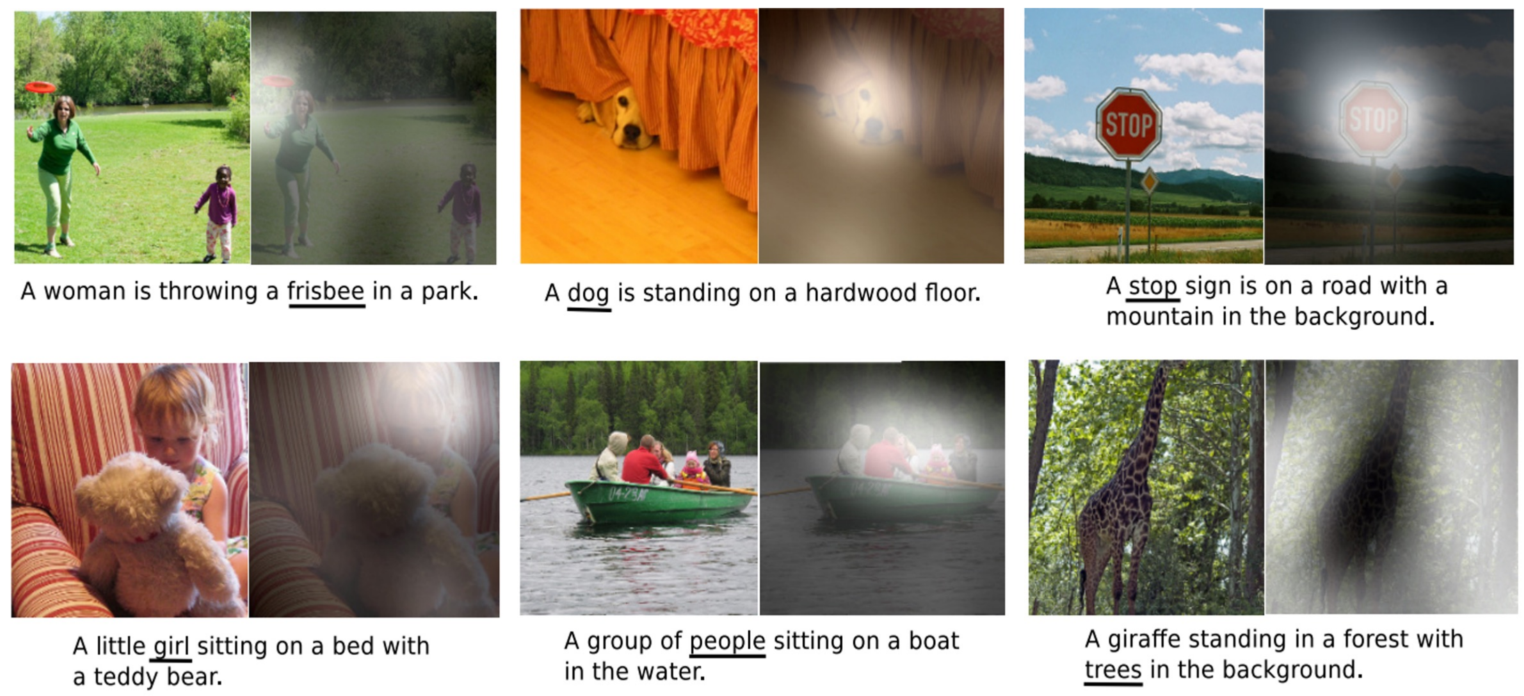
$$e_{t,i,j} = f_{\text{att}}(s_{t-1}, h_{i,j})$$

$$c_t = \sum_i a_{t,i,j} h_i$$

Each time step of decoder uses a different context vector that looks at different parts of the input image

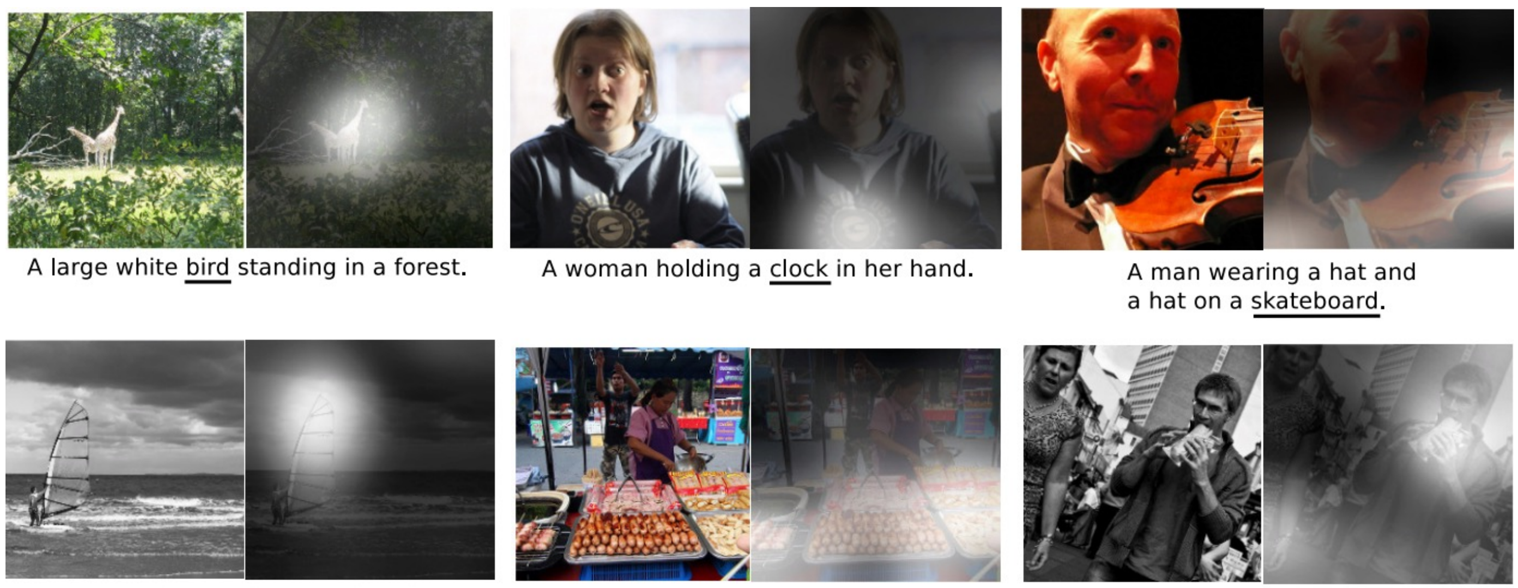
Example results

- Good captions



Example results

- Mistakes



A large white bird standing in a forest.

A woman holding a clock in her hand.

A man wearing a hat and a hat on a skateboard.

A person is standing on a beach with a surfboard.

A woman is sitting at a table with a large pizza.

A man is talking on his cell phone while another man watches.

Quantitative results

Dataset	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR
Flickr8k	Google NIC	63	41	27	-	-
	Soft-Attention	**67**	44.8	29.9	19.5	18.93
	Hard-Attention	**67**	**45.7**	**31.4**	**21.3**	**20.30**
Flickr30k	Google NIC	66.3	42.3	27.7	18.3	-
	Soft-Attention	66.7	43.4	28.8	19.1	**18.49**
	Hard-Attention	**66.9**	**43.9**	**29.6**	**19.9**	18.46
COCO	Google NIC	66.6	46.1	32.9	24.6	-
	Soft-Attention	70.7	49.2	34.4	24.3	**23.90**
	Hard-Attention	**71.8**	**50.4**	**35.7**	**25.0**	23.04

Source